Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/03/2005
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Project Title:	Development of a Gait Adaptability Training Program as a Countermeasure for Postflight Locomotor Dysfunction		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cohen, Helen (Baylor College of Medicine) Mulavara, Ajitkumar (NSBRI)		
Grant/Contract No.:	NCC 9-58-NA00403		
Performance Goal No.:			
Performance Goal Text:			
	Following their return to Earth, crewmembers experience disturbances in their ability to walk and maintain postural stability due to neural adaptation to the microgravity conditions of space flight. These changes can pose risks to crew safety and mission objectives if nominal or emergency vehicle egress is required immediately following long-duration spaceflight. At present, no operational countermeasure is available to mitigate these risks by facilitating rapid sensorimotor re-adaptation to 1-G. Therefore, the goal of this project is to develop an inflight balance and gait training program that will facilitate recovery of locomotor function after long-duration spaceflight. The proposed countermeasure is based on the concept of adaptive generalization. During this type of training the subject gains experience producing the appropriate adaptive motor behavior under a variety of sensory conditions and response constraints. As a result the subject learns to solve a class of motor problems, rather than a specific motor solution to one problem, i.e., the subject		

Task Description:	learns response generalizability or the ability to "learn to learn." under a variety of environmental constraints. Numerous studies support the concept that motor learning can be enhanced through training but to date practical applications have not been developed. The Gait Adaptability Training Program will be built upon the existing inflight treadmill exercise program approved for the International Space Station. The Gait Adaptability Program will be done simultaneously with nominal inflight treadmill exercise. By manipulating the sensory conditions of exercise by varying visual flow patterns, body load, and the stability of the walking surface we will systematically and repeatedly challenge the balance control system during treadmill walking. This training regimen will enhance sensorimotor adaptability, facilitating re-adaptation to unit (Earth) and partial (Mars) gravity after long-duration spaceflight. To develop the Gait Adaptability Training Program we are conducting a series of ground-based studies that will systematically investigate the training efficacy associated with exposing subjects to different combinations of sensory input during treadmill locomotion including alterations in visual flow, body loading, and support surface stability. We will also determine the optimal training schedule (distributed vs. massed) to maximize both the efficacy and efficiency of the training procedure. To determine the optimal training procedure and schedule, subjects are evaluated on completion of training on our previously developed Adaptive Functional Mobility Test (AFMT), which involves negotiating a complex obstacle course placed on medium density foam while wearing up/down reversing goggles. To perform this test subjects must rapidly reorganize their locomotor responses to compensate for the novel sensory distortions caused by the visuomotor transformation induced by the displacing goggles and the unstable support surface caused by the foam walking surface. The AFMT serves as a model system to evaluate ability to adaptively reorganize locomotor responses based on the preceding training program. We are also measuring trunk motion during each treadmill training session to characterize changes in dynamic postural stability during treadmill walking. Determining how locomotor stability changes during and between training sessions will allow us to determine the relationship between locomotor variability exhibited during training and the efficacy of each training mode to produce adaptive functional locomotor behavior as defined by performance on the AFMT. In the first year we have focused on development and validation of the virtual scenes and training periods used during the treadmill training program. Our results show that viewing simulated rotary self-motion during treadmill locomotion causes adaptive modification of sensorimotor integration in the control of position and trajectory during locomotion, which functionally reflects adaptive changes in the integration of visual, vestibular, and proprioceptive cues. We have confirmed that exposure to a rotating virtual visual scene during treadmill walking for a period of 20 minutes is sufficient to produce both strategic changes in dynamic postural stability during locomotion coupled with real plastic/adaptive changes in locomotor function. Both strategic and adaptive changes in locomotor function serve as the basis for the facilitating gait adaptation to new environments and these studies shows that optic flow can play a central role in this process and thus in locomotor adaptability training in general. This study is directly related to our current International Space Station study "Mobility" whose goal is to develop an inflight treadmill training program to increase adaptability of locomotor function after space flight. The information from the current NSBRI study will enable us to optimize the inflight training program and schedule. The results from this study will also be used to specify the requirements for future in-flight treadmills used during long-duration space flight. In the coming year we will continue work on using visual flow as a modality to increase gait adaptability by completing our comprehensive gait treadmill training study that is now in progress. We also hope to add support surface variation as a training modality by mounting a treadmill on a six-degree-of-freedom motion device and evaluating the efficacy of this type of training in conjunction with variation in visual flow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a clinical perspective exposure to varied sensorimotor stimuli during treadmill walking provides environmental enrichment to enhance gait and postural adaptability. A vast literature demonstrates that the central nervous system requires a rich and varied sensorimotor environment to maintain normal structure and function. Also, a converging body of evidence supports the use of environmental enrichment both before and after brain trauma, as an intervention to help restore behavioral function These studies indicate that exposure to environmental enrichment can play a central role in the restoration of functional mobility. Importantly, studies have shown that training comprised of challenging balance tasks compared to simple repetitive or treadmill exercises alone increase structural plasticity in the motor cortex and thalamus. Within the context of locomotor functional training and rehabilitation, it is clear that an enriched and challenging environment provided by virtual reality devices provides the optimal conditions to explore variability and in that exploration increase adaptability of motor function. Indeed, the concept of employing virtual reality technology as a tool for rehabilitation has become a topic of great interest among a number of researchers. A collateral benefit of the application virtual reality technology, in this context, will be to make gait training programs for both space flight and rehabilitation more interesting, ultimately leading to increased adherence to prescribed training regimens. In addition, the gait adaptability training program being developed by this project can be used to reduce the propensity of falling in the elderly. We are currently pursuing this line of research in collaboration with Dr. Ronita Cromwell and her graduate student Regina Buccello at the University of Texas Medical Branch in Galveston. This associated study will compare a group of elders receiving the specialized treadmill adaptability training with a group that receives the treadmill training only. Preliminary results are showing significantly improved gait function in elders that receive the training.
Task Progress:	1) Completed development of strategic and adaptive virtual visual scenes for use in treadmill training for Specific Aim 1. 2) Received IRB approval and completed a NASA required Test Rediness Review for the experimental hardware set-up. 3) Completed data collection for Specific Aim 1 and presented preliminary results at the Sensorimotor Team Retreat in Boston, July 20-22. 4) Completed installation and initial programming of the 6-degree-of freedom motion platform that will provide support surface variation during treadmill training for Specific Aim 2. 5) Finalized procedures and metrics for tests of adaptablity that serve to evaluate the proposed treadmill training programs.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Mulavara AP, Richards JT, Ruttley T, Marshburn A, Nomura Y, Bloomberg JJ. "Exposure to a rotating virtual environment during treadmill locomotion causes adaptation in heading direction." Exp Brain Res. 2005 Oct;166(2):210-9. PMID: 16034569 , Oct-2005

Articles in Peer-reviewed Journals	Nomura Y, Mulavara AP, Richards JT, Brady R, Bloomberg JJ. "Optic flow dominates visual scene polarity in causing adaptive modification of locomotor trajectory" Brain Res Cogn Brain Res. 2005 Dec;25(3):624-31. PMID: 16216478 , Dec-2005
Awards	N/A "NASA Performance Award" Jan-2004
Presentation	Bloomberg, J. J, Mulavara A. P., Peters, B. T., Cohen H. S., Richards J. T., Miller C. A., Brady R., Warren L. E. "Development of training programs to mitigate post space flight gait dysfunction" N/A Jun-2005
Presentation	Bloomberg, J. J. "Locomotor Dysfunction after Long-Duration Space Flight: Sensorimotor Issues and Training Interventions" N/A Aug-2005
Presentation	Richards J. T., Mulavara, A. P., Ruttley, T. M., Bloomberg, J. J. "The role of strategic changes in plastic adaptation of locomotor function" N/A Jun-2005